FTX3045S TEST ONE: SOLUTIONS & MARKING METHODOLOGY

NB: The memo is just a guideline; otherwise reasonable answers were accepted.

QUESTION 1

$$RCY = \left[\frac{P_3 + \text{Reinvested Coupons}}{P_0}\right]^{\frac{1}{n}} - 1$$ *[formula mark below but put it here]*

Determine P_0, this is the clean price

- Determining the dirty price.
  - Step 1: Value the bond as of the next coupon date (30/09/2024)
  - $\mathrm{P}_{NCD} = \frac{6.25\%}{2} \times R10{,}000 = R312.50$
  - $\text{Yield to maturity(r)} = \frac{10.994\%}{2} = 5.497\%$

- Price the bond as of the next coupon date

$$\mathrm{P}_{NCD} = \mathrm{R}312.50 + [R312.50 \times \frac{1-\left[\frac{1}{(1+5.497\%)^{23}}\right]}{5.497\%} + \frac{\mathrm{R}10{,}000}{(1+5.497\%)^{23}} \quad [8]$$

$$\mathrm{P}_{NCD} = R7{,}257.70$$

Markers:

- ✓ *Watch out for different methods – contact me if you are not sure.*
- ✓ *Here is the allocation of the 8 marks on the first part.*
  - *2 marks is for pricing the bond as of the next coupon date. Sometimes students don't mention that they are pricing the bond as of the next coupon date, but you can see it by looking at the first undiscounted coupon outside the bracket.*
    - *If a student fails to add the undiscounted coupon, then give 1 mark*
  - *1 mark for discounting the annuities and face value, make sure the n is correct (being 23) to get these 2 marks.*
  - *2 marks for the correct coupon amount of R312.50 (some students plug this into their formula $\frac{R625}{2}$ or some will do this $\frac{6.25\%}{2} \times R10{,}000$ – its fine you can give them the 2 marks. However, if they use the annual coupon of R625 give 1 mark.*
  - *2 marks for the correct semi-annual yield to maturity (YTM) of 5.497%. If a student uses 10.994% give 1 mark (This is a mark for being able to pick the correct YTM from the snapshot- The YTM should be used or written in full with all its decimals.*
  - *1 mark for the correct final of R7,257. (there will be minor differences because of rounding) Give full marks if answers is close but on condition that all the inputs are correct*

- Step 2: Discount that value to the settlement date to get the dirty price.
  - *The settlement date is given as 29th/08/2024*

$$P_{SD} = \frac{R7{,}257.70}{(1.05497)^{\frac{32}{183}}} = R7{,}190.10$$	(5)

Markers:

- *2 of these are **method marks** for discounting the value from step 1 (Price as of the next coupon date). What matters is they are using the value of the bond as of next coupon date from their first calculation.*
- *2 marks for getting the 32 days correct.*
- *1 mark for getting the 183 days correct.*

- Determine clean price = Dirty price – Accrued interest

  - $$\text{Annual Coupon} \times \frac{\text{Number of days from settlement date to next coupon date}}{365}$$

$$625 \times \frac{151}{365} = 258.56 \text{ [2]}$$

- ✓ *Markers*
  - *Carry forward the coupon amount from the previous calculation and give full 2 marks if everything else is correct. So the coupon amount does not have to be 625*
  - *Give 1 formula mark if the students wrote the formula and if everything else is incorrect.*
  - *Give 1 mark if everything else is correct except the 151days*
  - *If the final answer is far-off while the inputs are 100% correct you give 1 mark*
  - *Do not award any mark to students who use 366 days. Only award them 1 mark if they write the formula and correctly.*

- $\text{Clean price} = R7{,}190.10 - 258.56 = R6{,}931.54$ [2]

  - ✓ *Markers*
    - *The clean price calculation is a method mark for subtracting their own dirty price and their own accrued interest*

- Determine the selling price (P_1) = R10,000 (yield-to-maturity= Coupon rate) [2]
  - *Markers:*
    - ✓ *If they work it out and get the correct answer – give the full 2 marks.*
    - ✓ *However, do not give any part mark if they get the wrong answer by any margin (they did not need to calculate it in the first place). The answer must be exactly R10,000*

- Determine coupons and income from reinvesting the coupons

  - Coupon received: 312.50 (30th September 2024); 312.50 (31st March 2025), 312.50 (30th September 2025); 312.50 (31st March 2026)

  - $\text{Coupons and reinvestment income} = R312.50(1.045)(1.04)(1.04) + 312.50(1.04)(1.04) + 312.50(1.04) + 312.50 = R1{,}328.71$ [4]
    - *Markers*
      - ✓ *1 mark for each coupon compounded*

- ✓ *The coupon amount does not have to be correct, carry the coupon amount from the previous calculation.*
- ✓ *If they use annual coupons here but with correct compounding periods give 2 marks.*
- ✓ *If all the reinvestment rates are incorrect give but compounding periods are correct give 2 marks.*
- ✓ *Do not award a mark to students who discount the coupons.*

❖ <u>Determine the realized compound yield</u>

$$RCY = \left[\frac{P_3 + \text{Reinvested Coupons}}{P_0}\right]^{\frac{1}{n}} - 1$$ *[formula mark below]*

$$RCY = \left[\frac{\text{R}10{,}000 + \text{R}1{,}328.71}{\text{R}6{,}931.54}\right]^{\frac{1}{3.17}} - 1 = 16.76\%$$ **[4]**

$$16.76\% \times 2 = 33.53\%$$

- ✓ *If everything is wrong, check if the student wrote the formula and then give 1 mark formula. Don't give a formula mark if the student does not write the formula.*
- ✓ *Carry forward the student calculation for the clean price (denominators) and the total for coupons and reinvestment income - give full 4marks only if the compounding periods are correct $\frac{1}{3.17}$*
- ✓ *<u>If n(</u>$\frac{1}{3.17}$) is wrong but all the variables are positioned correctly, give 2 marks.*
- ✓ *You only give a formula mark if everything is wrong, but the student wrote the formula. Don't give a formula mark for students who does not write the formula*
- ✓ *In summary, a student who write a formula and then delivers the realised compound yield with a wrong n should walk away with a maximum of 2marks*

<u>QUESTION 2</u>

❖ Yes, it is true **[1]** *The key difference between these two bonds is the call feature.* <u>However, at high levels of interest rates relative to the coupon rate, a callable bond is unlikely to be called, hence investors will be indifferent between a callable and vanilla bond</u> **[1].** *Markers: To get these 2 marks, somewhere the student should respond to the question*

❖ The similar features include
- ✓ Positive convexity, **[1]**
- ✓ Since the bonds have the same coupon rate, at those levels they will trade at the <u>same price</u> and offer the <u>same return</u> **[1]**

- ***<u>Markers</u>***
  - ✓ *<u>Any of these two points</u>*
  - ✓ *<u>If they say both bonds will trade at a discount give 1 mark</u>*
  - ✓ *<u>Do not award to those who write (i) the price of both bonds will decrease or increase – that is a feature for all bonds, so the level of analysis should go deeper than that for a 3rd year Finance. (ii)both will offer high yields or low yields – no mark for this (iii) both bonds will mature at the same time</u>*

QUESTION 3

- Maximise returns
  - You generate returns by buying junk bonds that are less likely to default, such bonds have a higher yield, either through high coupons or they trade at a huge discount to the face value [1].
  - You can also generate returns by anticipating a change in the rating and then generate trading profit [1]
- You can generate trading profits in two ways.
  - If you anticipate a credit rating upgrade for a bond, buy the bond before the upgrade. When the bond rating is upgraded, the yield-to-maturity drops and the price increases – this allows you to sell the bond at a higher price for a capital gain [3].
  - Or if you anticipate a credit rating downgrade, you could sell the bond now if you had it in your portfolio or you can short sell it now. When the bond is downgraded the yield-to-maturity increases and the bond prices drops – this allows you to buy it back at a lower price for a capital profit [3].
    - *Markers: The question specifies "trading profits". So the strategy should generate profits from buying at a low price and selling at a high price, in either direction*
    - *You do not generate trading profits from earning a high coupon*
    - *Give them full 5 marks if they do this only*
      - *If you anticipate a credit rating upgrade for a bond, buy the bond before the upgrade. When the bond rating is upgraded, the yield-to-maturity drops and the price increases – this allows you to sell the bond at a higher price for a capital gain [3].*
      - *Or if you anticipate a credit rating downgrade, you could sell the bond now if you had it in your portfolio or you can short sell it now. When the bond is downgraded the yield-to-maturity increases and the bond prices drops – this allows you to buy it back at a lower price for a capital profit [3].*
    - *If they explain one of these trading strategies, then give 3 marks + 1 to make it 4; one mark for stating which is embedded in this explanation*

QUESTION 4

1. I would expect the South African reserve bank to increase the repo rate [1] to control inflation [1] *(watch out for this, this part can be embedded in the explanation below, do not forget to award the mark)*
   Explain how a repurchase rate (repo rate) hike controls inflation
   Controlling inflation:
   When the repurchase rate (repo rate) is increased, lending rates rise, making borrowing more expensive. As a result, citizens borrow less, reducing the money in circulation and lowering demand for goods and services. This, in turn, helps to reduce the demand-driven component of inflation [3] *(any explanation in this direction)*

2. If you are anticipating interest a repo rate hike you can generate trading profits by selling or short-selling short-term bonds. When the repo rate is hiked, short term

bonds yields will increase, and bond prices decrease. When the prices of short-term bonds decrease you can buy them back at a lower price, cover the short-sell and realise your capital gain.[3]

Markers

- ✓ *The question specifies "trading profits". So the strategy should generate profits from buying at a low price and selling at a high price or selling at a high price and buying back at a low price.*
- ✓ *You do not generate trading profits from earning a high coupon or rom buying a callable with a higher yield.*
- ✓ *For the second part, carry their prediction from the first part, if a student predicted a cut in the repurchase rate, which is wrong, they will lose marks on the first part but they can get full marks if their trading strategy is correct based on a repo rate cut*
    - *If you are anticipating interest a repo rate cut you can generate trading profits by buying short-term bonds. When the repo rate is cut, short term bonds yields will decrease, and bond prices increase. When the prices of short-term bonds increase you can sell them back at a higher price and realise a capital gain price and cover the short-sell.*[3]

QUESTION 5

- ❖ They did not cut the repurchase rate (repo rate) because they believe the risk of inflation remains high. Reducing the repo rate would increase the likelihood of inflation rising beyond the target band of 3-6%. A lower repo rate reduces lending rates, making borrowing cheaper and increasing disposable income, as citizens pay less on loans linked to the prime rate. With more money in people's hands, demand for goods and services rises, leading to higher prices and inflation. [2]
- ❖ They chose not to hike the repo rate to avoid further harming the economy, which showed signs of contraction in the first quarter of 2024 with a 0.1% decline. Raising the repo rate would negatively impact economic performance by making borrowing more expensive and increasing interest rates on existing loans linked to the repo and prime rates. This reduces disposable income, leading to decreased demand for goods and services, and consequently, lower production, which can contribute to further economic contraction [2]
- ❖ ***Markers:***
    - *Students were required to read the monetary policy statement to answer this question. So, its not a general question. To get full marks students should hint on the context of the two points: the context being that the economy is not doing well, it shrunk in the first uarter, and the inflation risk is still high. If they generalize their answer by sticking to theory of what happens when you hike or cut the repurchase rate then give a total of 2½ marks out of 4.*

QUESTION 6

1. Either of these two points
    - ✓ During recessions corporates may face financial challenges and default risk is relatively higher. Hence, corporates reduce or postpone dividend payments, which reduces their returns, while for bonds the payment of interest is obligatory which force issuers to maintain the interest payments. In recession bond are more likely to provide a fixed income, whereas corporate dividends can be withheld, which reduces returns [3].

✓ Given the high default risk and poor performance of corporations during recessions, investors tend to shift from equities to safer bonds. This shift drives down equity prices, reducing their returns, while increasing bond prices and boosting their returns. [3].

2. The similarity is that in both cases you have to find the present value of the future cashflows [1]. The difference: If you are pricing a bond at the coupon date, you compounding periods are a whole number and all the remaining coupons belong to the buyer[1]. However, if you are pricing a bond between coupon dates; you must adjust for accrued interest and the compounding periods will include a fraction of a period [1].

   Markers
   - ✓ *On the similarity, do not award a mark for answers like in both cases you use the yield-to-maturity, its too basic*
   - ✓ *For the difference, they should give both sides, if they give one side then give 1 mark.*

QUESTION 7

1. Yield curves
   - ✓ Upward sloping and inverted yield curve [1].
   - ✓ Upward sloping predicts a boom, inverted yield curve predicts a recession [1]
   - ✓ ***Markers:*** *You can allocate a ½ if one part is correct*
2. The yield curve reflects market sentiment; hence it shows the position that market participants take when they are anticipating a certain state of the economy. If the market participants are correct, then the yield curve can predict economic conditions.[2]
3. One possible reason for an upward sloping curve
   - ✓ The market might be anticipating interest rates to increase in future. The higher interest expectations will be incorporated into the spot rates resulting in higher 5-year yields than 1-year yields [2]. *Markers: the key word here is "future" interest rates to increase (not just interest rates to increase)*
   - ✓ The market could be dominated by short term investors who are demanding higher returns for longer term bonds [2].
   - ✓ Maybe there is high demand and low supply for short-term bonds – leading to higher prices and lower yields, while on long-term bonds we could be experiencing high supply and low demand – leading to relatively low prices and higher yields. [2]

     Markers: *do not award marks for students who write the names of the theories without explaining how theory explain the shape of the curve. For example some students write only this: Pure expectations theory, market segmentation theory, liquidity preference theory*

   One possible reason for an inverted yield curve
   - ✓ The market might be anticipating interest rates to decrease in future. The lower interest expectations will be incorporated into the spot rates resulting in higher 5-year yields than 1-year yields [2]. *Markers: the key word here is "future" interest rates to decrease (not just interest rates to decrease)*
   - ✓ The market could be dominated by long term investors who are demanding higher returns or a liquidity premium for short term bonds [2].

- ✓ Maybe there is high demand and low supply of long-term bonds leading to to higher prices and lower yields, while short-term bonds could be experiencing high supply and low demand – leading to relatively low prices and higher yields. [2]
  Markers: *do not award marks for students who write: Pure expectations theory, market segmentation theory, liquidity preference theory*

QUESTION 8

1. It is the **relationship between the yield-to-maturity and time to maturity** of a set of **uniform/similar bonds.** [1]
   Markers
   - ✓ *For the definition to be correct all the bolded and underlined points should be included in the definition. If one of them is missing them the answer is wrong.*
   - ✓ *Do not award a mark to this "It is the relationship between the yield-to-maturity and time to maturity for bonds" If the bonds are not similar or uniform then its not a term structure of interest rates.*
2. The yield-to-maturity of a zero coupon do not have exposure to re-investment risk**[2],** this makes it possible for the term structure of interest rates to show how bond yields change with time only (time premium), without the yields being contaminated by reinvestment risk. [1]
   Markers
   - ✓ *Do not give a mark to those who only write "zero coupon bonds do not pay coupons"; This is not complete, and it does not answer the question, it just lays the foundation for the answer. They need to specify that the "the yield-to-maturity of a zero coupon do not have exposure to re-investment risk"*

QUESTION 9

1. Any two of these (1 mark for each)

- There is an inverse relationship between bond price and yields [1]
- For **small changes in yields**, the price sensitivity of a given bond is roughly the same-for an increase/decrease in yield *(Markers: the bold and underlined point need to be included to get a mark here)*
- For **large changes in yields,** the price sensitivity is not the same for an increase and a decrease in the required yield. *(Markers: the bold and underlined point need to be included to get a mark here)*
- For **large changes in yields** for a given bond, the percentage price increase is greater than the price decrease. *(Markers: the bold and underlined point needs to be included to get a mark here)*
- The price sensitivity is not the same for all bonds.

2. Calculations

a. <u>Modified duration</u>

$$D^* = \frac{D}{(1+r)}$$

$$D^* = \frac{10}{(1+9\%)} = 9.17\text{[2]}$$

✓ *<u>Markers</u>*

- *2 marks if all variables are correct*
- *1 mark if numerator is wrong but the denominator is correct*
- *If the denominator is wrong, do not award any mark*

b. <u>Money duration</u>

$$\text{MoneyDur} = D^* \times \text{PV}^{Full}$$

$$\text{Bond price} = \frac{\text{R1,000}}{(1+9\%)^{10}} = R422.41\text{[1]}$$

$$\text{MoneyDur} = 9.17 \times R422.41 = R3{,}873.51\text{[2]}$$

✓ *<u>Markers</u>*

- *Carry the modified duration from the previous calculation and award 2 marks. As long as they multiply their modified duration with a price, give 2 marks. We awarded a mark for the price, so they can still get these 2 marks even if the price is wrong*

c. Price value of a basis point for this bond

$$\text{PVBP} = \text{Money duration} \times 0{,}01\%$$

$$\text{PVBP} = \text{R3,873.51} \times 0{,}01\% = \text{R0.39 [2]}$$

✓ *<u>Markers</u>*

- *Carry the money duration from the previous calculation and award 2 marks. As long as they multiply their money duration with the price, give 2 marks.*

<u>Markers</u>: Some students will do this but it's not worth it because of the few marks allocated (the answers will differ because of round

$$\text{PVBP} = \frac{(\text{PV}_-) - (\text{PV}_+)}{2}$$

$$\text{PVBP} = \frac{422.7987 - 422.0235}{2} = 0.39$$

d. Assuming interest rates increase by 2.3%, estimate the price change for this bond using duration estimate.

$$\frac{\Delta P}{P} \approx -D^* \times \Delta y$$

$$\frac{\Delta P}{P} \approx -9.17 \times 2.3\% = -0.21\text{[2]}$$

✓ ***Markers***

- *Carry the modified duration from the previous calculation and award 2 marks. If they multiply their modified duration with the change in yield, give 2 marks.*
- *The negative sign accounts for the inverse relationship between prices and yields. If the student fails to capture the negative sign, do not award any mark. However, give the full 2 marks if the student does specifies that the bond prices will "decrease" after doing this (9.17x2.3%= -0.21 or the student can say the price will decrease by 0.21. [I emphasised this in the last lecture]. No mark if a student simply writes does 9.17x2.3%=21.091%*
- *If they write the formula correctly and get everything wrong then give 1 mark*